TU Ilmenau, Fakultät für Informatik und Automatisierung
FG Algorithmik
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TECHNISCHE UNIVERSITÄT
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Algorithms – Exercise Sheet 3

Discussion on Monday, 2026-05-18

Exercise 1 (n-th Complex Roots of Unity)

For $n \geq 1$, the n-th complex roots of unity are the solutions of the equation $z^n = 1$, i.e. the following complex numbers:

$$\omega^0 = 1, \quad \omega^1, \quad \omega^2, \quad \omega^3, \quad \ldots, \quad \omega^{n-1}; \qquad \text{where } \omega \text{ is defined as } \omega := \exp(2\pi i/n) = \cos(2\pi/n) + i \cdot \sin(2\pi/n).$$

It is useful to think of these numbers as being specified using polar coordinates in the complex plane: they are n equally spaced points on the unit circle.

(a) Plot these numbers on the unit circle for $n = 1$, $n = 2$, $n = 3$, $n = 4$ and $n = 16$.

(b) Show that for any $k \in \mathbb{Z}$, the sum of the k^{th} powers of these numbers is 0 if n does not divide k, and n otherwise:

$$\sum_{\ell=0}^{n-1} (\omega^\ell)^k = \begin{cases} 0 & \text{if } n \text{ does not divide } k, \\ n & \text{otherwise.} \end{cases}$$

(c) Show that the product of these numbers is 1 if n is odd and -1 if n is even:

$$\prod_{\ell=0}^{n-1} \omega^\ell = \begin{cases} +1 & \text{if } n \text{ is odd,} \\ -1 & \text{if } n \text{ is even.} \end{cases}$$

Exercise 2 (FFT: Multiplication of two Polynomials)

In this exercise we multiply the two polynomials $A(x) = 2x + 1$ und $B(x) = 3x^2 - 4x + 1$ by using the (Inverse) Fast Fourier-Transformation (FFT / IFFT).

(a) Determine the correct n for this polynomial multiplication and give the two tuples $(a_0,\ldots,a_{n-1})$ and $(b_0,\ldots,b_{n-1})$.

How can you choose the correct n in general?

(b) Execute the necessary calls of the FFT and IFFT-algorithm by writing down the values of all variables during the execution. A call to the FFT algorithm evaluates a polynomial — for each recursive call, write down the polynomial and the points at which the polynomial is being evaluated.

(If you like, you can use the schema shown in Fig. 2. You find an example for the filled-out schema in Fig. 1.)

(c) Give the coefficients of the result polynomial $C(x) = A(x) \cdot B(x)$.

Check your result by computing the coefficients with normal multiplication of the polynomials.

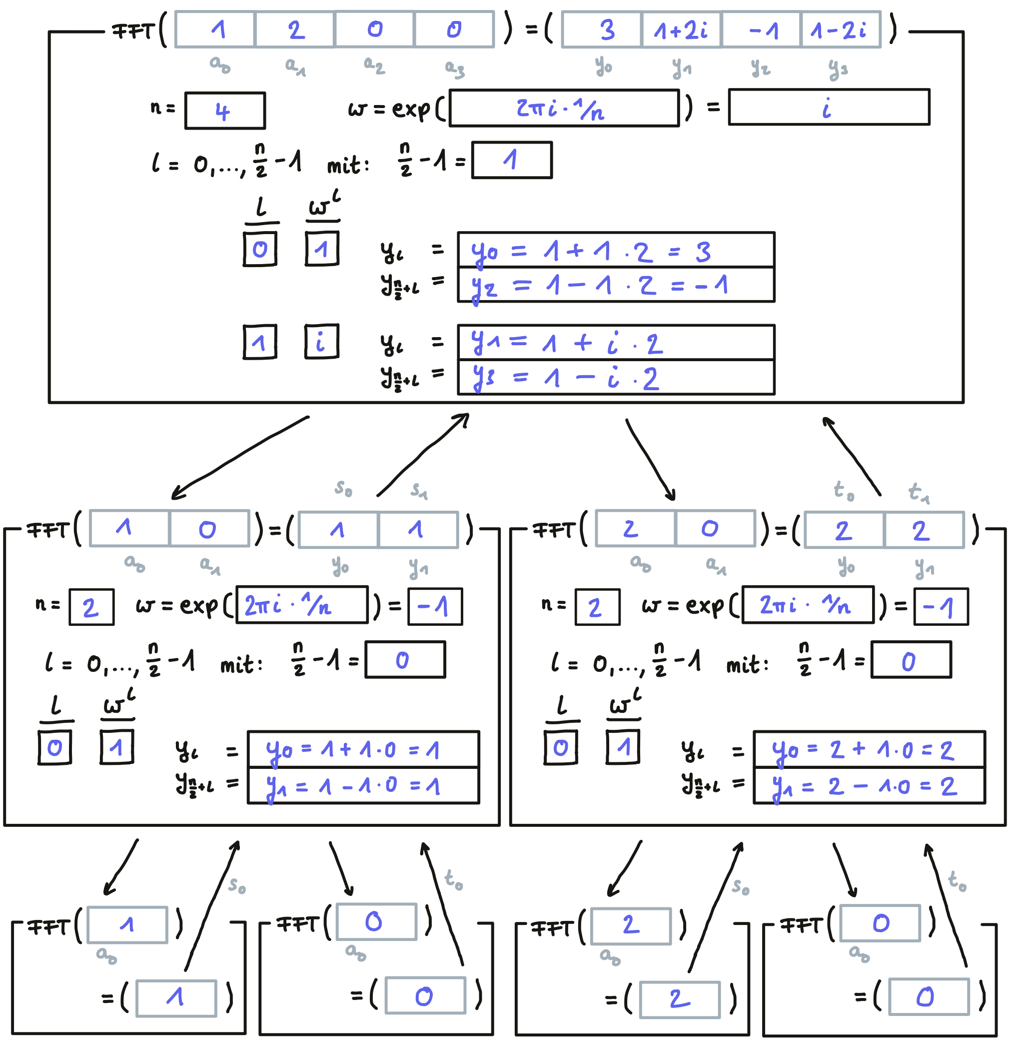

Figure 1: An example execution of the FFT algorithm with the proposed schema. Here, the FFT algorithm has been executed on the coefficient tuple $(a_0,\ldots,a_3) = (1,2,0,0)$, this corresponds to the polynomial $0x^3 + 0x^2 + 2x + 1$. The result is the tuple $(y_0,\ldots,y_3) = (3, 1+2i, -1, 1-2i)$, which are the values of the polynomial at the four positions $(1, \omega, \omega^2, \omega^3)$ for the primitive 4th root of unity $\omega = i$ (which means $\omega^2 = -1$ and $\omega^3 = -i$).

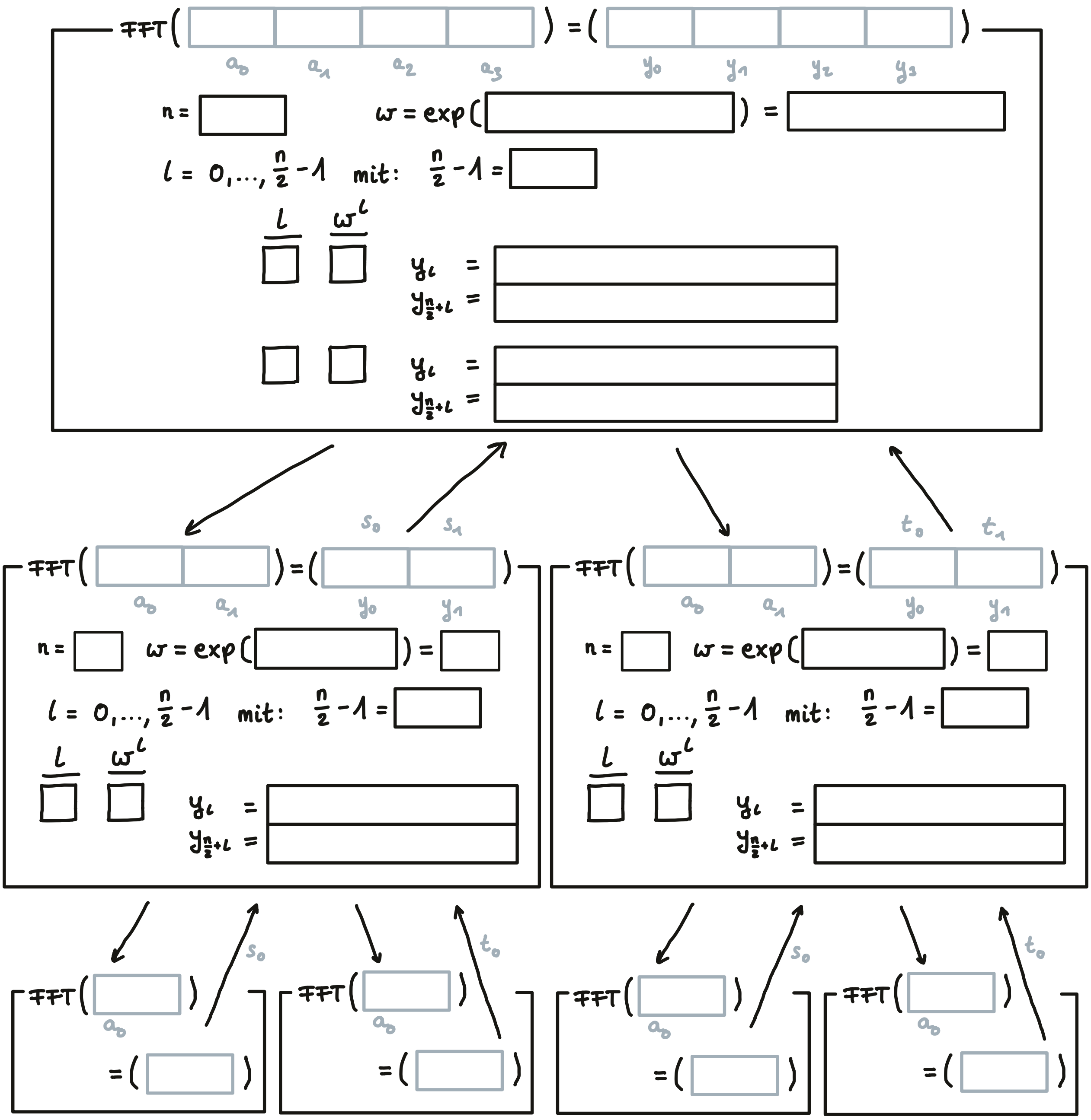

Figure 2: Schema für manuelle Ausführung des FFT-Algorithmus